

Revised Proposal for Modeling and Simulation

Proposed Title

Modeling the Rising Poverty Threshold and the Growth of Newly Poor Households in Pasay City Using Time-Series Regression and Compartmental Dynamics

Reason for Revision

The original proposed project is no longer practical to complete under the current situation. It was initially intended as a two-person project, but it is now being completed individually. In addition, the original concept depended heavily on 3D modeling work, which has become a major bottleneck in terms of time, technical production, and overall feasibility. Because of these constraints, the original proposal can no longer be completed at a quality level appropriate for submission.

A revised topic is therefore proposed that remains within the scope of modeling and simulation but is more feasible, data-driven, and grounded in a real-world problem visible in the local environment. The revised study focuses on poverty vulnerability in Pasay City and nearby urban communities, where rising living costs and economic pressure can be observed directly in everyday life.

Background

The Philippine Statistics Authority (PSA) conducts the Family Income and Expenditure Survey (FIES) nationwide every three years as the country's main source of household income and expenditure data (Philippine Statistics Authority [PSA], n.d.-b). According to PSA, poverty incidence among families in the Philippines fell to 10.9 percent in 2023 from 14.2 percent in 2021, while poverty incidence among the population fell to 15.5 percent in 2023 from 18.1 percent in 2021 (PSA, n.d.-a). At the same time, the World Bank notes that the pandemic partly reversed earlier poverty reduction gains and that high inequality continues to limit upward mobility in the Philippines (World Bank, 2025).

Problem Statement

Many urban households that were previously stable are becoming economically vulnerable as the cost of living increases over time. This creates two connected problems: first, the poverty threshold or effective minimum income needed to meet basic needs tends to rise; second, more lower-middle and middle-income households risk falling below that threshold and becoming part of the "new poor." The proposed study aims to model both processes mathematically using publicly available household income, poverty, and inequality data (PSA, n.d.-a; PSA, n.d.-b; World Bank, 2025).

General Objective

To develop a mathematical and statistical model that explains and predicts how the poverty threshold changes over time and how those changes increase the risk of middle-income households in Pasay City becoming newly poor (PSA, n.d.-a; World Bank, 2025).

Specific Objectives

- To analyze historical poverty and household income indicators from official Philippine datasets, particularly PSA poverty statistics and Family Income and Expenditure Survey data (PSA, n.d.-c; PSA, n.d.-b)
- To build a time-series model for estimating the upward movement of the poverty threshold over time using available macroeconomic and poverty indicators (PSA, n.d.-a; World Bank, 2025)
- To build a compartmental transition model that represents how economically vulnerable households move from middle-income status to newly poor status under rising cost pressure (World Bank, 2025)
- To simulate future scenarios showing how continued increases in living costs may affect the number of newly poor households in an urban setting such as Pasay City (World Bank, 2025; PSA, n.d.-a)

Conceptual and Mathematical Framework

The study will combine two modeling components.

1. Poverty Threshold Trend Model

The first component treats the poverty threshold as a time-dependent quantity:

$$PT(t) = \alpha + \beta_1 t + \beta_2 X_t + \varepsilon_t$$

where PT(t) is the estimated poverty threshold at time t, and X_t may include explanatory indicators such as inflation-related or income-distribution variables derived from public datasets (PSA, n.d.-a; World Bank, 2025).

2. Household Transition Model

The second component models the movement of households between economic groups using a compartmental framework:

dM/dt	=	-βMP	+	γR
dP/dt	=	βMP	-	δP
dR/dt = δP - γR				

where M represents economically vulnerable middle-income households, P represents newly poor households, and R represents households that temporarily recover or stabilize. In this framework, the transition parameter β captures the effect of worsening economic conditions on vulnerable households (World Bank, 2025).

This combined approach is appropriate because the study does not only ask whether poverty exists, but also how a rising economic threshold can gradually pull more households into poverty over time (World Bank, 2025; PSA, n.d.-a).

Data Sources

Dataset	Use in the study
PSA Poverty Statistics	Historical poverty incidence, poverty among families, and official poverty indicators (PSA, n.d.-a; PSA, n.d.-c)
PSA Family Income and Expenditure Survey (FIES)	Household income, expenditure, education, employment, and income-group patterns (PSA, n.d.-b; PSA, n.d.-d)
World Bank Philippines poverty and inequality indicators	Long-run inequality and poverty-related context for calibration and interpretation (World Bank, 2025)

These sources are suitable for Python-based analysis because they are provided through official statistics portals or downloadable data pages that can be converted into CSV or Excel workflows for pandas-based processing (PSA, n.d.-a; PSA, n.d.-b; World Bank, 2025).

Methodology

The study will follow a quantitative modeling and simulation approach. First, relevant datasets will be collected, cleaned, and transformed into a usable format in Python. Second, descriptive statistical analysis will be performed to identify trends in poverty incidence, household income, and vulnerability indicators (PSA, n.d.-b; PSA, n.d.-a).

After the descriptive phase, a time-series regression model will be fitted to estimate the trend of the poverty threshold or equivalent poverty-pressure indicator over time. A compartmental simulation model will then be calibrated to represent transitions of households from middle-income to newly poor conditions under different parameter settings. Numerical simulation will be performed to test baseline, moderate-stress, and high-stress scenarios (World Bank, 2025).

The outputs will be limited to tables, line graphs, and simulation plots. No 3D assets or 3D modeling pipeline will be required, making the study significantly more feasible within the available time frame.

Expected Output

The study is expected to produce the following outputs:

- A cleaned Python-ready dataset from official public sources (PSA, n.d.-a; PSA, n.d.-b)
- Trend graphs showing poverty-related indicators over time (PSA, n.d.-a)
- A predictive model of the rising poverty threshold or poverty-pressure trend (World Bank, 2025; PSA, n.d.-a)
- A compartmental simulation showing how vulnerable middle-income households may become newly poor over time (World Bank, 2025)
- Scenario-based interpretations relevant to urban households in Pasay City

Significance of the Study

The revised topic is socially relevant because it reflects a real and visible urban issue: families that were once economically stable may gradually become newly poor as living costs rise and opportunities become more uncertain. The World Bank emphasizes that inequality, uneven opportunity, and pandemic-related shocks continue to affect upward mobility in the Philippines (World Bank, 2025). This makes the study meaningful not only as a mathematical exercise but also as a realistic social science application of modeling and simulation (World Bank, 2025).

The topic is also academically appropriate because it uses mathematical modeling, statistics, simulation, and data analysis without requiring 3D content production. This makes the revised proposal more feasible for an individual student project while still meeting the analytical goals of a modeling and simulation course.

Request for Approval

In view of the major implementation challenges of the original topic, a revision of the proposal is respectfully requested. The revised study preserves the core idea of modeling and simulation, but shifts the implementation toward a more feasible, data-driven, and socially grounded problem that can be completed individually within the available timeline.

References

Philippine Statistics Authority. (n.d.-a). *Full Year Poverty Statistics*. PSA OpenSTAT. https://openstat.psa.gov.ph/PXWeb/pxweb/en/DB/DB_1E_FY/?tablelist=true

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World Bank. (2025). *Poverty headcount ratio at national poverty lines (% of population) - Philippines*. World Bank Poverty and Inequality Platform. <https://data.worldbank.org/indicator/SI.POV.NAHC?locations=PH>